

PROJECT AND TEAM INFORMATION

Project Title

SmartShield:Rule-BasedCreditCardFraudDetectionUsingDataStructures

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Credit card fraud is a serious problem in digital payment systems because unauthorized transactions can cause financial loss and reduce user trust. Detecting suspicious transactions quickly is therefore important for improving transaction security. SmartShield will develop a rule-based credit card fraud detection system using basic Data Structures and Algorithms instead of Artificial Intelligence or Machine Learning. The system will analyse transaction amount, time, merchant and location to identify unusual behaviour. The project will demonstrate how Hash Tables, Deques, Sorting and Graphs can be combined to detect suspicious transaction patterns in a simple, transparent and understandable way.

State of the Art / Current solution

Modern banking systems monitor transactions using automated fraud detection methods. These systems analyse transaction amount, location, merchant behaviour, transaction frequency and historical spending patterns. Many real-world solutions use predefined rules along with statistical, Artificial Intelligence or Machine Learning techniques. These methods are suitable for processing large amounts of financial data. SmartShield will use a simpler rule-based approach for academic demonstration. Fundamental DSA concepts will be implemented to organise transaction data and identify suspicious activity without using AI or ML.

Project Goals and Milestones

The main goal is to develop a functional terminal-based credit card fraud detection system that demonstrates practical use of DSA concepts.

Phase 1:

Planning & Requirement Analysis

Define problem statement, objectives, scope, requirements, and select C++/Data Structures.

Phase 2:

System Design & Development

Design system architecture, flowchart, database/data handling, and implement fraud detection using C++ and Data Structures.

Phase 3:

Testing & Final Implementation

Test the system with different transactions, identify fraud cases, fix errors, generate reports, and prepare final documentation/demo.

Project Approach

The Credit Card Fraud Detection System will be developed using C++ and Data Structures. C++ will be used for implementing the complete system, while Data Structures will be used for efficient storage, processing, searching, and management of transaction records.

Structures and Classes will be used to store transaction information such as Transaction ID, amount, date, time, location, and transaction status. Arrays or vectors will be used to store multiple transaction records. A queue will be used to process incoming transactions in the order in which they are received. A linked list can be used to maintain dynamically changing transaction records, while a stack can be used to maintain recent transaction history.

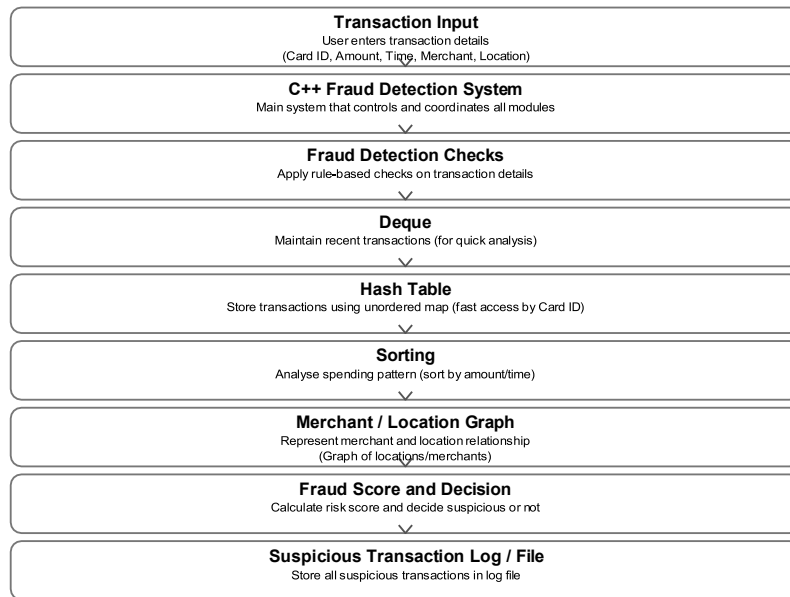
Searching algorithms will be used to find specific transactions, while sorting algorithms will be used to arrange transactions according to amount, time, or risk score. Hashing can be used for fast searching of Transaction IDs.

Pointers will be used for dynamic data structures such as linked lists.

C++ functions will divide the system into different modules such as transaction input, transaction processing, fraud detection, searching, sorting, and report generation. File handling will be used to store and retrieve transaction records and fraud-detection results.

The fraud detection algorithm will analyze transaction characteristics and identify suspicious transactions based on predefined conditions or a risk score. Finally, the system will classify each transaction as Legitimate or Fraudulent and display the result.

System Architecture (High Level Diagram)



Rule-based C++ system using Deque, Hash Table, Sorting and Graphs for transaction analysis

Assumptions

The project assumes that every transaction contains a valid card ID, amount, time, merchant and location. Historical transactions are assumed to represent the normal spending behaviour of the cardholder. Location information is assumed to be sufficient for identifying geographically distant transactions. A fraud score of two or more will be treated as suspicious. The system is intended for academic demonstration and is not a production banking fraud detection system.

References

1. *Thomas H. Cormen et al., Introduction to Algorithms, MIT Press.*
2. *Bjarne Stroustrup, The C++ Programming Language, Addison-Wesley.*
3. *cppreference, C++ Standard Library and Container Documentation.*
4. *GeeksforGeeks, Data Structure sand Algorithms in C++.*
5. *Course material and lecture notes on Data Structures and Algorithms.*